

Use Case 1: Courier Offering System Modernization

Context

Your company operates a delivery platform across 15 countries with 50,000+ active couriers. The legacy offering system sends delivery opportunities to couriers, but it's monolithic and inflexible.

Business Requirements

The business needs to:

- A/B test different offer presentations
- Show personalized earnings calculations based on courier tier, location, time of day, and promotions
- Roll out features gradually (by city, courier segment, or percentage)
- Support different earning models (flat rate, distance-based, surge pricing, tips prediction)

Current State

- Single offer template hardcoded in the backend
- Earnings calculation scattered across 3 services (pricing, promotions, assignments)
- Any change requires full deployment across all regions
- No experimentation capability
- Response time SLA: 200ms (p95), currently at 180ms
- 2M offers sent per hour during peak times

The Challenge

The revamp has started, but critical decisions are pending. The team is split between two approaches:

Approach A: Build a "template engine" that renders offers server-side with a DSL for defining offer variants

Approach B: Move to a configuration-driven approach where the backend sends raw data and mobile apps handle presentation logic

Examples of Assessment Areas - You can (and should) cover more

1. System Design:

- Design the end-to-end architecture for the modular offering system

2. Technical Decision Making:

- Evaluate Approach A vs B - what's your recommendation and why?

3. Technical Leadership:

- The mobile team is concerned about approach B increasing their complexity. How do you facilitate this decision?

4. Migration Strategy:

- What metrics would you track to ensure the migration is successful?

Use Case 2: Team Guidance - Real-time Fraud Detection

Context

A product team of 4 engineers (2-3 years experience each) is building a real-time fraud detection system for suspicious courier behavior patterns. The Staff Engineer role is to provide technical guidance without doing the work for them.

Detection Requirements

The team needs to detect patterns like:

- Couriers marking deliveries as complete far from the destination
- Suspicious earnings patterns (collusion with merchants)
- Fake GPS data

Current Problems

Architecture confusion: They chose Apache Kafka + Flink but are struggling to implement the streaming pipeline. Processing delays are at 45 seconds instead of target <5 seconds

Data quality issues: They're getting incomplete location data and don't know how to handle it

Testing difficulties: No clear strategy for testing the fraud detection rules

Scope creep: Started with 3 fraud patterns, now stakeholders want 12, and the team keeps saying "yes"

Knowledge gaps: Team has strong backend experience but limited streaming/ML knowledge

Recent Developments

- Sprint review is in 3 days, and they have nothing to demo
- One engineer wants to "start over with a simpler approach"
- Product manager is frustrated and escalating to leadership
- Team morale is low

Important point: You are not expected to take the Tech Manager role but to work with the Tech Manager to solve the problem

Examples of Assessment Areas - You can (and should) cover more

1. Diagnostic & Assessment:

- What questions would you ask the team to understand the root causes?

2. Technical Guidance Strategy:

- How would you guide them without solving the problem yourself?

3. Immediate Action Plan:

- What would you do in the next 3 days before the sprint review?

4. Long-term Team Development:

- How would you structure knowledge transfer on streaming systems?

5. Working with the Tech Manager and Principals:

- How would you work with the tech manager and principals to address the technical issues?

The Presentation

You will have 1 hour to go over both cases. So please, manage your time.

It would be advised to have a presentation with your insights, system design, and all the solutioning you came up with and thought about.

Please, make sure you address all the requirements.

Feel free to reach out to Gilberto if you need any assistance.

Good luck!

Thanks!